

Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: BLURB | Context: EU/French Pharmaceutical Regulatory NLP — BLURB Assessment

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

BLURB's task taxonomy (NER, PICO, relation extraction, sentence similarity, document classification, QA) is explicitly scoped to PubMed-based biomedical applications [Q46, Q48], with clinical and 'other high-value verticals' deferred to future work [Q46, Q197]. The deployment requires categories for regulatory document genres (SmPCs, PILs, CTD modules, patent claims) and regulatory-specific tasks such as cross-document consistency checking and regulatory equivalence STS — none of which are represented in BLURB's taxonomy. The QA scope is further restricted to yes/no and three-way classification [Q49, Q51, Q124], excluding the kind of structured field-extraction or compliance-verification task structure relevant to regulatory submissions. Web-confirmed: no public regulatory NLP benchmark with these task categories exists for French [WEB-12, WEB-23], reinforcing the structural gap. The taxonomy does include NER, document classification, and STS at an abstract level, which are the surface task types the deployment also uses, so the gap is one of category specificity rather than total absence.

ID	Evidence
Q46	"We focus on PubMed-based biomedical applications, and leave the exploration of the clinical domain, and other high-value verticals to future work." (p.6)
Q48	"BLURB is comprised of a comprehensive set of biomedical NLP tasks from publicly available datasets, including named entity recognition (NER), evidence-based medical information extraction (PICO), relation extraction, sentence similarity, document classification, and question answering." (p.7)
Q49	"For question answering, prior work has considered both classification tasks (e.g., whether a reference text contains the answer to a given question) and more complex tasks such as list and summary [42]." (p.7)
Q51	"For simplicity, we focus on the classification tasks such as yes/no question-answering in BLURB, and leave the inclusion of more complex question-answering to future work." (p.7)
Q124	"For the two-way (yes/no) or three-way (yes/maybe/no) question-answering task, the encoding is similar to the sentence similarity task." (p.12)
Q197	"Future directions include: further exploration of domain-specific pretraining strategies; incorporating more tasks in biomedical NLP; extension of the BLURB benchmark to clinical and other high-value domains." (p.22)
WEB-12	No regulatory-expert-annotated pharmaceutical NLP dataset (French or English) identified; DART Italian corpus does not use regulatory-expert annotators (https://arxiv.org/html/2510.18475)
WEB-23	Practitioner review confirms NLP for EU SmPC/prescribing information remains hybrid with no validated regulatory-specific benchmark (https://intuitionlabs.ai/articles/nlp-regulatory-labeling)

Strengths

- BLURB's six-task structure (NER, RE, document classification, STS, QA, PICO) names the surface task types the deployment also relies on (NER, STS, document classification), providing a generic abstraction layer [Q48, Q195]
- The benchmark explicitly acknowledges that prior biomedical evaluations use heterogeneous task selections [Q37], and offers a stable task-grouped scoring scheme [Q52, Q150] that the deployment could in principle adopt for internal evaluation

ID	Evidence
Q37	"In contrast, prior work on biomedical pretraining tends to use different tasks and datasets for downstream evaluation, as shown in Table 2." (p.6)

Q48	“BLURB is comprised of a comprehensive set of biomedical NLP tasks from publicly available datasets, including named entity recognition (NER), evidence-based medical information extraction (PICO), relation extraction, sentence similarity, document classification, and question answering.” (p.7)
Q52	“To compute a summary score for BLURB, the simplest way is to report the average score among all tasks. However, this may place undue emphasis on simpler tasks such as NER for which there are many existing datasets. Therefore, we group the datasets by their task types, compute the average score for each task type, and report the macro average among the task types.” (p.7)
Q150	“The BLURB score is the macro average of average test results for each of the six tasks (NER, PICO, relation extraction, sentence similarity, document classification, question answering).” (p.14)
Q195	“To facilitate this study, we create BLURB, a comprehensive benchmark for biomedical NLP featuring a diverse set of tasks such as named entity recognition, relation extraction, document classification, and question answering.” (p.21)

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories for the deployment include regulatory NER (INNs, ATC codes, posology, excipients, EMA-template qualifiers), regulatory document classification (SmPC vs PIL vs CTD genre), regulatory equivalence STS (SmPC↔PIL, original↔revised), and cross-document consistency checking. None of these categories is present in BLURB; the closest analogs are generic biomedical NER, BIOSSES-style STS, and HoC-style abstract classification.

IO-2: Yes — BLURB omits regulatory document genres and regulatory-specific task types entirely. The benchmark explicitly defers 'other high-value verticals' to future work [Q46, Q197]. No public French or EU regulatory NLP benchmark covering these categories was identified [WEB-12, WEB-23].

IO-3: BLURB's PICO task (clinical trial P/I/O extraction) [Q59, Q111] and the molecular biology entity types in JNLPBA (DNA, RNA, cell line, cell type) [Q58] are not directly relevant to the regulatory submission deployment; they consume evaluation weight in the macro-averaged BLURB score [Q52, Q150] without contributing to regulatory-task signal.

IO-4: Content validity is harmed because the benchmark's task taxonomy underrepresents the regulatory-document task constructs the deployment needs to evaluate (genre-specific NER, regulatory equivalence, cross-document consistency) and includes research-oriented categories (PICO, molecular-biology NER) that do not generalize to SmPC/PIL/CTD prose.

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Q46	“We focus on PubMed-based biomedical applications, and leave the exploration of the clinical domain, and other high-value verticals to future work.” (p.6)
Q48	“BLURB is comprised of a comprehensive set of biomedical NLP tasks from publicly available datasets, including named entity recognition (NER), evidence-based medical information extraction (PICO), relation extraction, sentence similarity, document classification, and question answering.” (p.7)
Q52	“To compute a summary score for BLURB, the simplest way is to report the average score among all tasks. However, this may place undue emphasis on simpler tasks such as NER for which there are many existing datasets. Therefore, we group the datasets by their task types, compute the average score for each task type, and report the macro average among the task types.” (p.7)
Q58	“JNLPBA. The Joint Workshop on Natural Language Processing in Biomedicine and its Applications shared task [27] is a NER corpus on PubMed abstracts. The entity types are chosen for molecular biology applications: protein, DNA, RNA, cell line, and cell type. Some of the entity type distinctions are not very meaningful. For example, a gene mention often refers to both the DNA and gene products such as the RNA and protein. Following prior work that evaluates on this dataset [34], we ignore the type distinction and focus on detecting the entity mentions. We use the same train, development, and test splits as in Crichton et al. [14].” (p.8)

Q59	“EBM PICO. The Evidence-Based Medicine corpus [44] contains PubMed abstracts on clinical trials, where each abstract is annotated with P, I, and O in PICO: Participants (e.g., diabetic patients), Intervention (e.g., insulin), Comparator (e.g., placebo) and Outcome (e.g., blood glucose levels). Comparator (C) labels are omitted as they are standard in clinical trials: placebo for passive control and standard of care for active control. There are 4300, 500, and 200 abstracts in training, development, and test, respectively.” (p.8)
Q111	“Conceptually, evidence-based medical information extraction is akin to slot filling, as it tries to identify the PIO elements in an abstract describing a clinical trial.” (p.11)
Q150	“The BLURB score is the macro average of average test results for each of the six tasks (NER, PICO, relation extraction, sentence similarity, document classification, question answering).” (p.14)
Q197	“Future directions include: further exploration of domain-specific pretraining strategies; incorporating more tasks in biomedical NLP; extension of the BLURB benchmark to clinical and other high-value domains.” (p.22)
WEB-12	No regulatory-expert-annotated pharmaceutical NLP dataset (French or English) identified; DART Italian corpus does not use regulatory-expert annotators (https://arxiv.org/html/2510.18475)
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Information Gaps

- Whether any internal EMA tooling (Scientific Explorer, PICROSS, EurEKA) defines a public task taxonomy compatible with BLURB's framing

Requires Expert Verification

- The full set of regulatory task categories required for ANSM-compliant submission review (e.g., posology consistency, contraindication scope verification) — would benefit from regulatory affairs expert specification

Remediation

Gap 1: BLURB's task taxonomy excludes regulatory document genres and regulatory-specific tasks (regulatory NER, regulatory equivalence STS, cross-document consistency).

Recommendation: Build a complementary task suite covering SmPC/PIL/CTD NER, regulatory equivalence STS (SmPC↔PIL, original↔revised), and EMA-template compliance classification. Use BLURB's six-task structure [Q48, Q52] as a reusable scoring scaffold but redefine the underlying task categories against EMA QRD v10.4 [WEB-1, WEB-2].

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Q48	“BLURB is comprised of a comprehensive set of biomedical NLP tasks from publicly available datasets, including named entity recognition (NER), evidence-based medical information extraction (PICO), relation extraction, sentence similarity, document classification, and question answering.” (p.7)
Q52	“To compute a summary score for BLURB, the simplest way is to report the average score among all tasks. However, this may place undue emphasis on simpler tasks such as NER for which there are many existing datasets. Therefore, we group the datasets by their task types, compute the average score for each task type, and report the macro average among the task types.” (p.7)
WEB-1	EMA QRD SmPC Template v10.4 (Feb 2024) — defines the formulaic template language the deployment must process (https://www.ema.europa.eu/en/human-regulatory-overview/marketing-authorisation/product-information-requirements/product-information-qrd-templates-human)
WEB-2	https://www.ema.europa.eu/en/documents/template-form/qrd-product-information-template-version-104-highlighted_en.pdf